2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Objective

This experiment is divided into four simulations:

1. Periodic and Aperiodic Signals
2. Verification of Even and Odd Signal Decomposition
3. Amplitude and Time Scaling of Signals
4. Time Shifting and Time Reversal of Signals

Program:

```
% Program to Generate Periodic and Aperiodic Signals


clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
% Periodic Signal
subplot(2,1,1);
plot(t, x1, 'b', 'LineWidth', 2);
grid on;
title('Periodic Signal');
xlabel('Time (s)');
```

```
ylabel('Amplitude');
xlim([-5 5]);
ylim([-1.2 1.2]);
% Aperiodic Signal
subplot(2,1,2);
plot(t, x2, 'r', 'LineWidth', 2);
grid on;
title('Aperiodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
xlim([-5 5]);
ylim([0 1.2]);
```

Expected Output:

When you run the MATLAB code, you should observe two plots:

1. Periodic Signal (Sinusoidal Wave):

o Characteristics: The first subplot shows a sinusoidal waveform that repeats its pattern every 1 second, as it is a 1 Hz sinusoid.

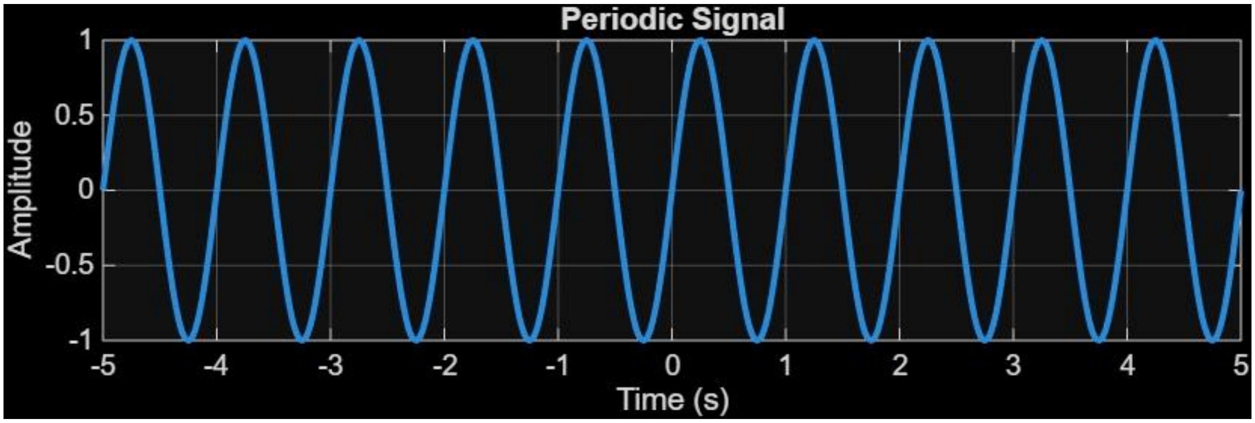
o Description: The signal oscillates between -1 and 1, completing two full cycles over the 2-second duration.

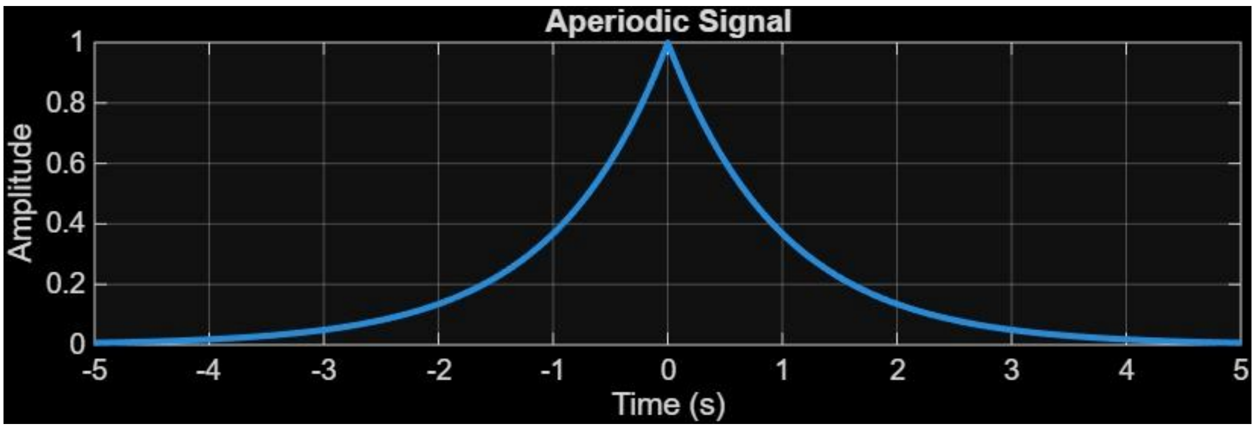
2. Aperiodic Signal (Exponentially Decaying Signal):

o Characteristics: The second subplot shows an exponentially decaying signal. Unlike the sinusoidal signal, this signal does not repeat its pattern.

o Description: The signal starts at amplitude 2 and decays towards zero as time progresses, reflecting its aperiodic nature.

Output Waveform:





Result:

- Periodic Signal: The sinusoidal signal repeats itself at regular intervals, making it periodic.
- Aperiodic Signal: The exponentially decaying signal does not repeat, showcasing that it is aperiodic.
- This test case illustrates the key differences between periodic and aperiodic signals bygenerating and visually analysing both types using MATLAB.

Experiment 2.2 Verification of Even and Odd Signal Decomposition

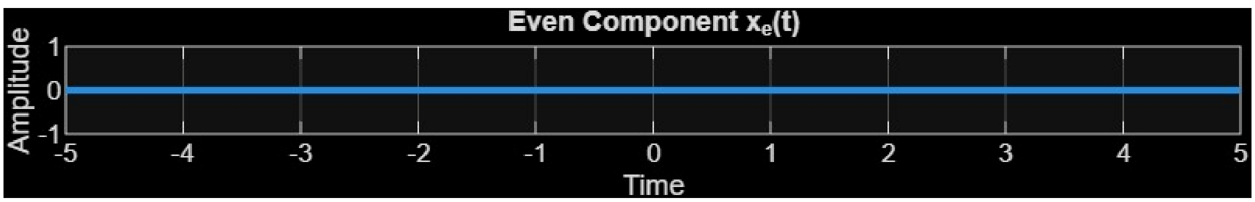
Program

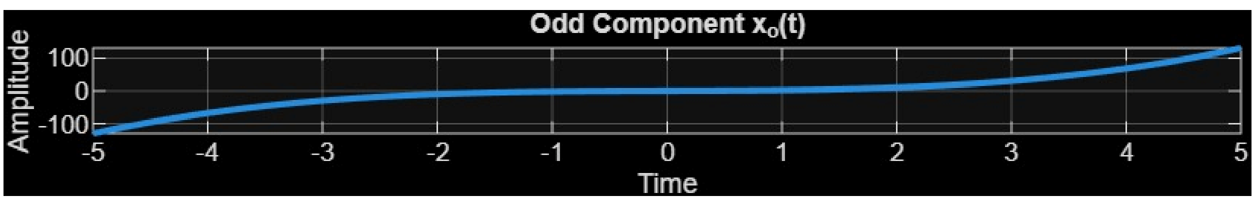
```
% Verification of Even and Odd Signal Decomposition

clc;

clear;

close all;

% Time axis

t = -5:0.1:5;

% Original Signal

x = t.^3 + t;

% Time-Reversed Signal x(-t)

x_rev = (-t).^3 + (-t);

% Even Component

xe = (x + x_rev)/2;

% Odd Component

xo = (x - x_rev)/2;

% Reconstruction of Original Signal

x_reconstructed = xe + xo;

% Plotting

figure;

% Original Signal

subplot(4,1,1);

plot(t, x, 'b', 'LineWidth', 2);
```

```
grid on;

title('Original Signal x(t)');

xlabel('Time');

ylabel('Amplitude');

xlim([-5 5]);

% Even Component

subplot(4,1,2);

plot(t, xe, 'r', 'LineWidth', 2);

grid on;

title('Even Component x_e(t)');

xlabel('Time');

ylabel('Amplitude');

xlim([-5 5]);


% Odd Component

subplot(4,1,3);

plot(t, xo, 'g', 'LineWidth', 2);

grid on;

title('Odd Component x_o(t)');

xlabel('Time');

ylabel('Amplitude');

xlim([-5 5]);

% Reconstructed Signal
```

```
subplot(4,1,4);

plot(t, x_reconstructed, 'm', 'LineWidth', 2);

grid on;

title('Reconstructed Signal x_e(t) + x_o(t)');

xlabel('Time');

ylabel('Amplitude');

xlim([-5 5]);
```

Output Waveform:





Result

Signal equals sum of even and odd parts.